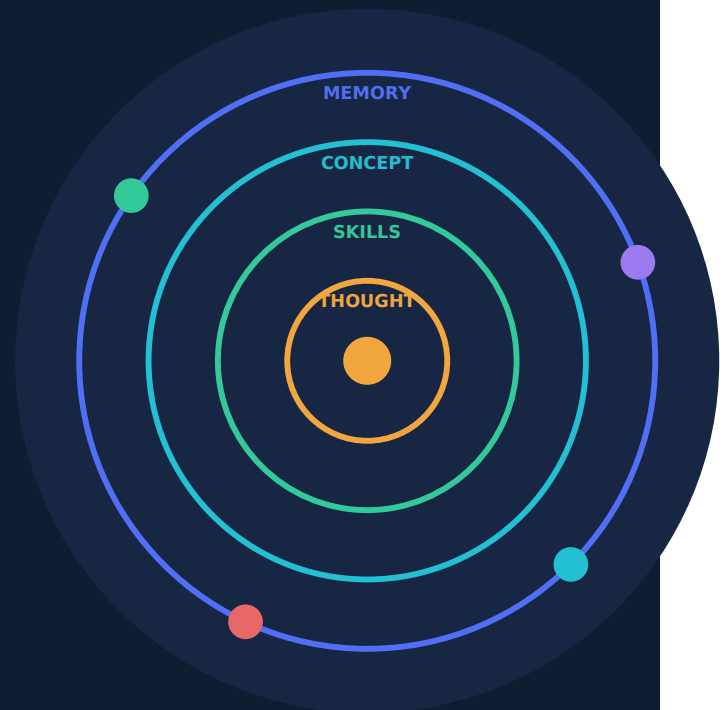


DENDRITRON-ARM

Memory, Concept Geometry, and Recurrent Reasoning

Exact 2/3-gram memory, layer-2 concept alignment, shared LoRA skills, experts, branches, and DeepLoop



BLOCK 1 -> BLOCK 2 -> NEXT ROUND

Technical review brief | Revision 2.0 | 3 August 2026

Architecture synthesis and implementation: Ryan Carson

Original Dendritron notes and the CPU/ARM objective are credited as the starting point.
This revision documents Carson's technical synthesis, extensions, and implementation boundary.

Contents

Revision 2.0 follows the mechanism from exact surface memory into concept geometry, reusable operations, expert-owned branches, and recurrent thought. Research citations appear beside the mechanism they support; the final pages preserve full references and source scope.

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TECHNICAL READING PATH

The shortest reading path is **Sections 4-8**: DeepSeek 2/3-gram memory -> LNGram and JTD concept retrieval -> collision handling -> the integrated universal/skill/expert/branch diagram -> operation-space hierarchy.

1. Why this architecture exists

Dendritron-ARM begins with one primary constraint: production training and reasoning belong on CPU/ARM. A large donor model constructs frozen semantic assets once, offline. The live recipient then spends memory bandwidth on sparse rows and compute on selected skills, experts, branches, and two repeatedly visited physical blocks.

ARCHITECTURAL THESIS

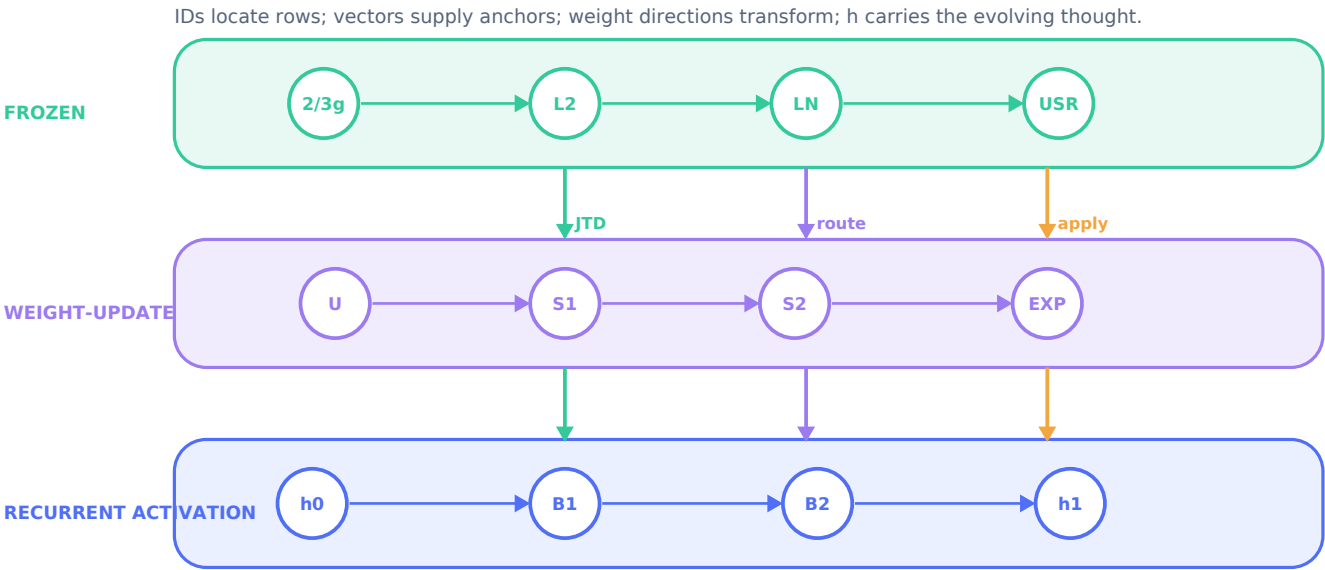
Remembering a stable phrase and developing a thought are separate computational jobs. Conditional memory stores stable semantic reference; the two-block recurrent state carries changing reasoning.

Decision	Reason	Consequence
Offline donor states	Reuse semantic structure already learned by a large model.	Dense donor encoding is paid once; runtime uses sparse lookup.
2/3-gram + definition memory	Phrase composition and word senses provide complementary granularity.	Local knowledge reaches the recipient before recurrent reasoning.
Universal -> skill -> expert	Common directions, reusable operations, and specialization learn at different scales.	Small adapters can serve many high-dimensional experts.
Expert-owned branches	Competing deductions, causes, contrasts, and interventions need explicit identities.	Each candidate returns movement, evidence, residual, and trace.
Two physical blocks	Parameter storage and sequential thought depth become separate axes.	Stored block depth is 2; effective depth is 2R.
Fixed 25/75	Stable reference receives one share; changing conditional computation receives three.	Memory and reasoning scale together under one invariant.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
DeepSeek Engram [1] separates static lookup from dynamic computation. Memory Grafting [2] shows that frozen donor hidden states can populate exact conditional memory.	Carson combines those ideas with a fixed layer-2 concept frame, reusable LoRA skills, high-dimensional expert junctions, typed branches, a fixed 25/75 law, and CPU/ARM-first execution.

2. The architecture keeps its spaces separate

The model uses several related coordinate systems, each with a distinct job. Surface IDs locate rows. Memory vectors supply semantic anchors. JTD aligns views. Weight-space directions define transformations. The 2,048D activation carries the current thought.



Space	Contents	Function
Surface / address	Raw Qwen IDs, UTF-8 spans, canonical suffix tuples, punctuation boundaries	Exact identity, lookup, and trace
Frozen memory	Layer-8/24 phrase rows and layer-2 definition rows	Stable knowledge anchors
Latent address	LNGram multi-route discrete symbols and concept cells	Repeatable addresses derived from live hidden states
Joint concept	Fixed layer-2 definition geometry plus JTD-aligned views	Euclidean neighborhood and sense retrieval
Operation	Universal basis, shared skill LoRAs, user coefficients, expert parameters	How the state may change
Thought / trace	Evolving h plus row, sense, skill, expert, branch, and source pointers	Continuity and auditability across rounds

TYPE DISCIPLINE

A definition vector remains a concept location in activation space. A universal direction remains a basis element in LoRA weight-update space. A word or token ID remains an address and output symbol.

3. Why layers 2, 8, and 24 were chosen

The hidden states move semantic learning into a one-time construction stage. Each stored vector is a complete 2,048D BF16 donor state; spelling, frequency, row identity, definition text, and provenance remain attached as metadata.

Asset	Extraction	Architectural role
Bigram / trigram L8	Final non-padding token of the isolated phrase at donor layer 8	Earlier compositional phrase view
Bigram / trigram L24	Same phrase and final token at donor layer 24	Deeper composed phrase view
Definition sense L2	Concept-marker readout at hidden_states[2] for every retained definition sense	Frozen lexical concept anchor; polysemous senses remain separate
Recipient state	Initial working width 2,048	Direct bridge to donor assets and simple JTD maps

The final-token rule

For a causal donor, the last token of an isolated phrase has already incorporated every earlier token in that phrase. One phrase therefore produces one complete state per selected layer:

$$e^{(l)}(s) = F_{Qwen}(s)_{last}^{(l)}, \quad l \text{ in } \{8, 24\} \quad | \quad d_q = F_{Qwen}(definition_q + Concept)_{marker}^{(2)}$$

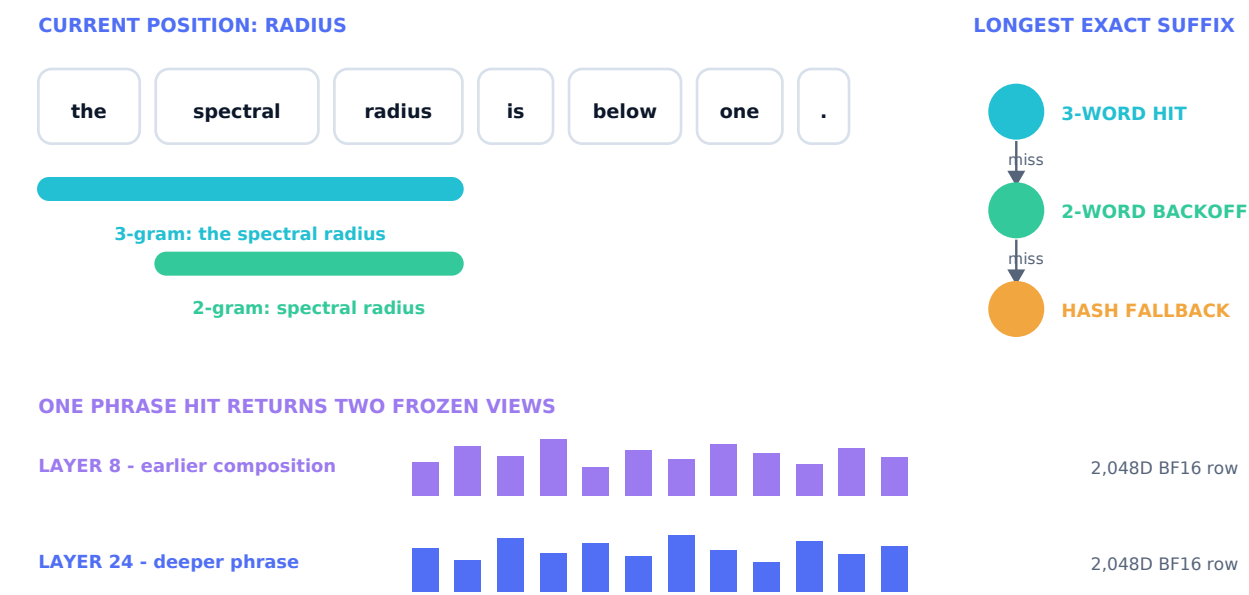
MECHANISM VERSUS DONOR

DeepSeek supplies the Engram conditional-memory mechanism. Qwen3.5-35B-A3B supplies Carson's frozen phrase values. Qwen layer 2 supplies the definition concept anchors.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Memory Grafting [2] reports its strongest tested grafting result with Qwen3.5-35B-A3B and uses layers 8 and 24 as early/middle and deeper donor views.	Carson keeps both full views for every selected bigram and trigram, then aligns them to a separate layer-2 definition frame rather than fusing them during extraction.

4. DeepSeek 2/3-gram memory and Carson's grafted banks

DeepSeek Engram treats the local suffix ending at each token as a deterministic memory address. Its core uses compressed token IDs, 2-gram and 3-gram orders, several hash heads, context-aware gating, and host-side prefetch. Memory Grafting adds an exact table of frequent phrases whose values are frozen donor states. Carson combines the two designs.



Carson's concrete memory build

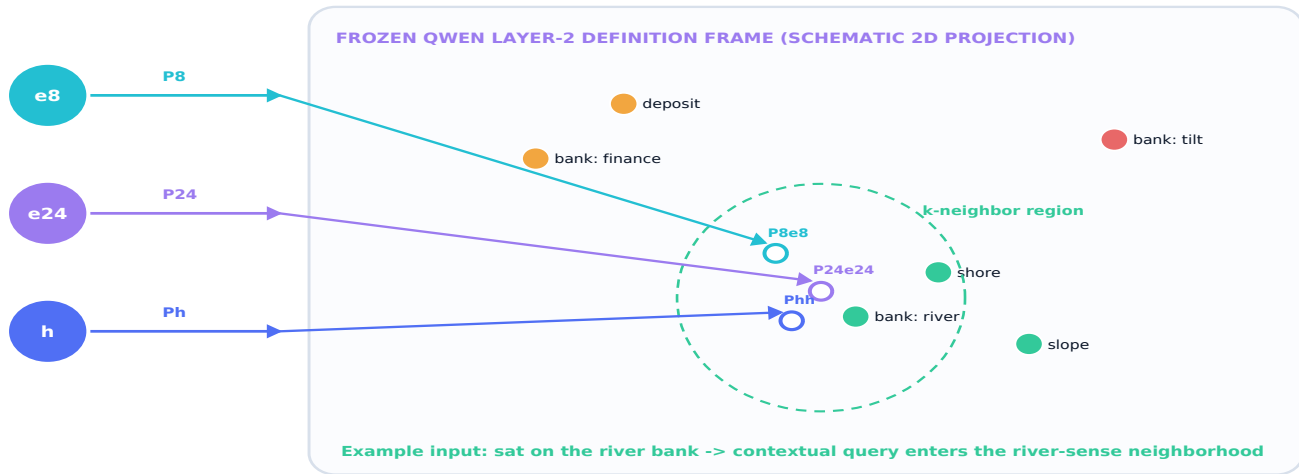
- A 200-million-token science/math-heavy corpus produced **500,000 bigrams** and **500,000 trigrams** after the content-word filter.
- Each selected surface phrase retains its text, frequency, row identity, and two full Qwen donor values: layer 8 and layer 24.
- At position t, the addressor checks the complete three-word suffix first, then the complete two-word suffix. This is longest exact matching.
- A three-word hit supplies greater specificity; a two-word hit supplies broader coverage for stable local relations and entities.
- An exact phrase miss activates multi-head Hash-Engram coverage. The current complete word still exposes its definition-sense candidates, producing the full 3 -> 2 -> 1 route.
- Raw punctuation remains in the live Qwen stream. Canonical punctuation classes remain available to the Hash-Engram address path.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Engram [1] found a strong fixed-budget reference with {2,3}-grams and reported a small regression after adding 4-gram capacity. Lngram [3] likewise found joint 2/3-gram retrieval strongest overall. Memory Grafting [2] supplies exact longest-suffix donor rows plus hash fallback.	Carson narrows the frozen inventory to the top 500k of each useful order, concentrates it on science/math language, stores two Qwen views per phrase, and routes exact 3 -> 2 before the trainable hash fallback.

5. From Engram addresses to layer-2 concept neighbors

Three mechanisms cooperate here. Surface Engram answers which exact phrase occurred. LNGram converts the evolving hidden state into a repeatable latent concept address. JTD moves phrase and live views into the fixed layer-2 definition geometry, where Euclidean k-nearest-neighbor retrieval selects the active concept senses.

SOURCE VIEWS



The complete concept-resolution path

- **Surface candidates.** Exact 3/2-gram lookup returns $e^{(8)}$ and $e^{(24)}$; the current word returns every retained layer-2 sense d_q .
- **LNGram address.** $\text{RMSNorm}(h)W_q$ is thresholded into several short bit routes; consecutive route symbols form exact latent 2/3-gram IDs. Repeated hidden patterns therefore land in repeatable concept cells.
- **JTD alignment.** $P_8e^{(8)}$, $P_{24}e^{(24)}$, and P_hh enter the layer-2 reference frame. Every definition anchor d_q stays fixed.
- **Layer-2 neighbor retrieval.** Candidate senses and nearby LNGram-linked concepts are ranked by Euclidean distance. Contextual phrase views and the live query jointly weight the selected neighbors.
- **Return movement.** Signed concept movement maps back into the 2,048D thought state and supplies skill/expert routine evidence.

$$q_t = \text{Fuse}(P_h h_t, g_8 P_8 e^{(8)}, g_{24} P_{24} e^{(24)}, r_{LN})$$

$$N_k(q_t) = \underset{s \in C_t}{\text{argmin}}^{(k)} ||q_t - d_s^{(2)}||_2$$

TWO MEANINGS OF L2

Layer-2 means the donor's second hidden-state readout. The subscript 2 on the distance means the Euclidean norm. JTD brings the input to the concept anchors; the anchors retain their original donor locations.

RESEARCH BASIS

LNGram [3] supplies learned binary multi-route symbols and exact latent 2/3-gram lookup. Locality Preserving Joint Transfer [4] supplies the idea of domain-specific projections into a shared frame while preserving k-neighbor structure.

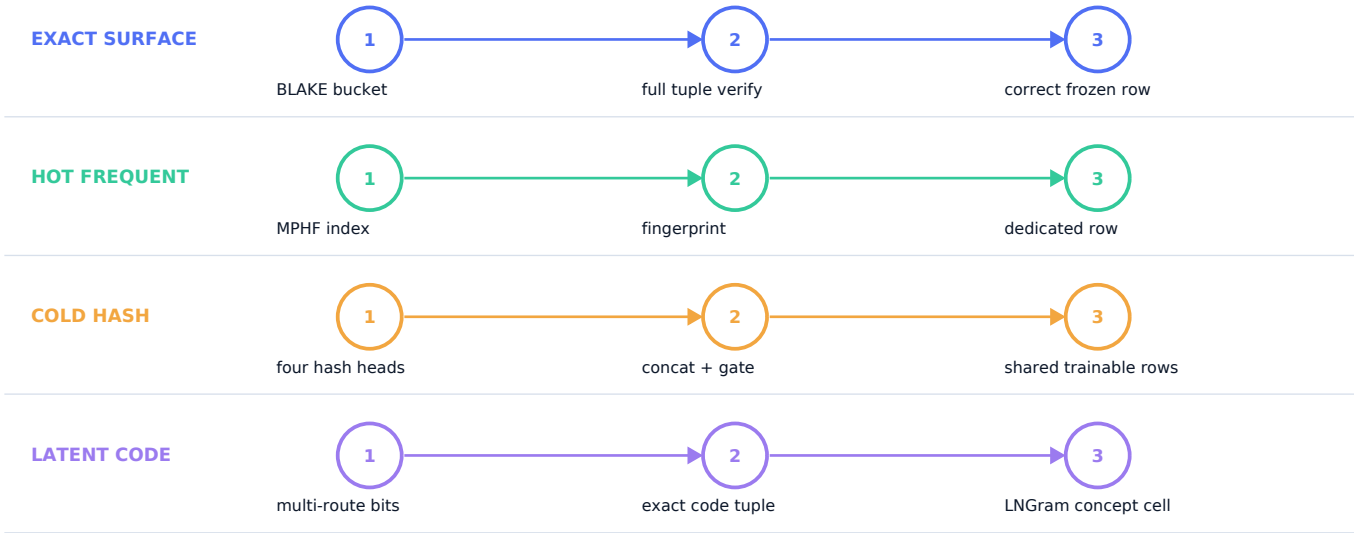
DENDRITRON-ARM EXTENSION

Carson fixes Qwen layer-2 definition senses as semantic landmarks, learns separate maps for layer-8, layer-24, and live recipient states, and uses their Euclidean neighborhoods to resolve concepts and route later computation.

6. Collision handling across the memory system

The word collision names several different events. Dendritron assigns each event its own resolution rule, telemetry, and evidence path. Exact phrase rows, hash fallback rows, LNGram codes, and user experience records therefore retain separate namespaces.

ONE WORD, FOUR DISTINCT COLLISION QUESTIONS

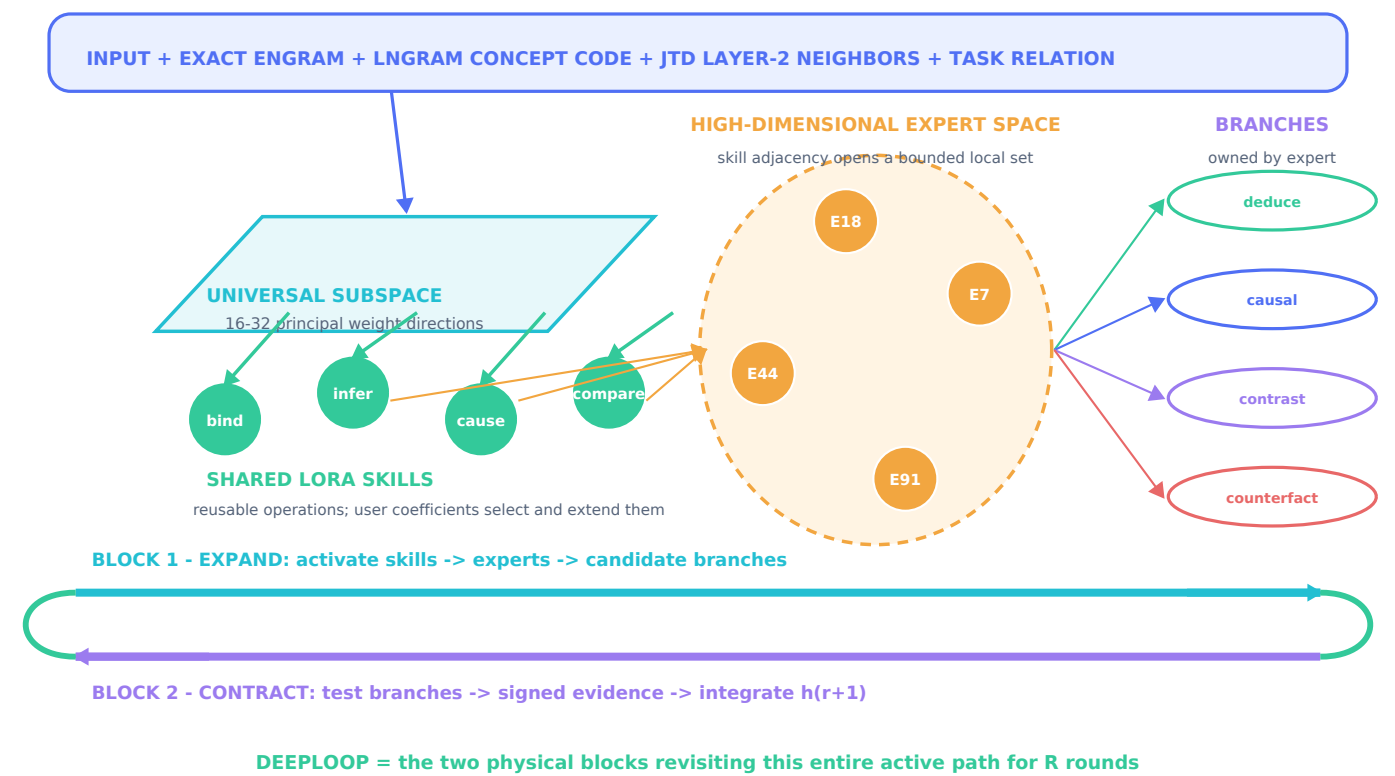


Collision type	Resolution	Meaning
Surface bucket	BLAKE2b-128 narrows the bucket; the complete canonical Qwen-ID or word tuple is compared before row return.	Compact indexing with exact identity
Canonical duplicate	All matching surface forms keep provenance; deterministic frequency/row policy selects the active frozen row.	Several labels may point to one canonical address
Hash-Engram sharing	Several independent hash heads, concatenation, and contextual gates distribute collision risk.	Long-tail parameter sharing and regularization
LNGram alias	Multiple short bit routes and optional subtables provide several latent views; occupancy and neighborhood quality are measured.	Quantized concept equivalence
User experience key	The latest verified commit becomes active for the exact user/task/context key; earlier versions retain trace and rollback identity.	Versioned experience rather than accidental sharing

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
The collision-free hot-tier study [9] uses MPHF plus fingerprints for frequent keys and multi-head hashing for the cold tier. Its controlled results show mixed validation gains from collision removal alone and highlight gating-credit dynamics. Extinction Depth [17] offers an adjacent finite-state analogy for tracking unresolved symbolic collisions across extensions.	Dendritron gives frequent exact grafted phrases dedicated rows, preserves multi-head hashing as trainable miss coverage, verifies compact surface buckets with full tuples, and records route-stratified gate/loss telemetry before changing the collision policy.

7. The complete task -> skill -> expert -> branch -> loop architecture

This figure places every active component in one view. Concept evidence and the task relation enter a low-dimensional universal foundation, activate shared LoRA skills, open a local high-dimensional expert neighborhood, and instantiate expert-owned branches. DeepLoop surrounds this execution path because both physical blocks revisit the hierarchy every round.



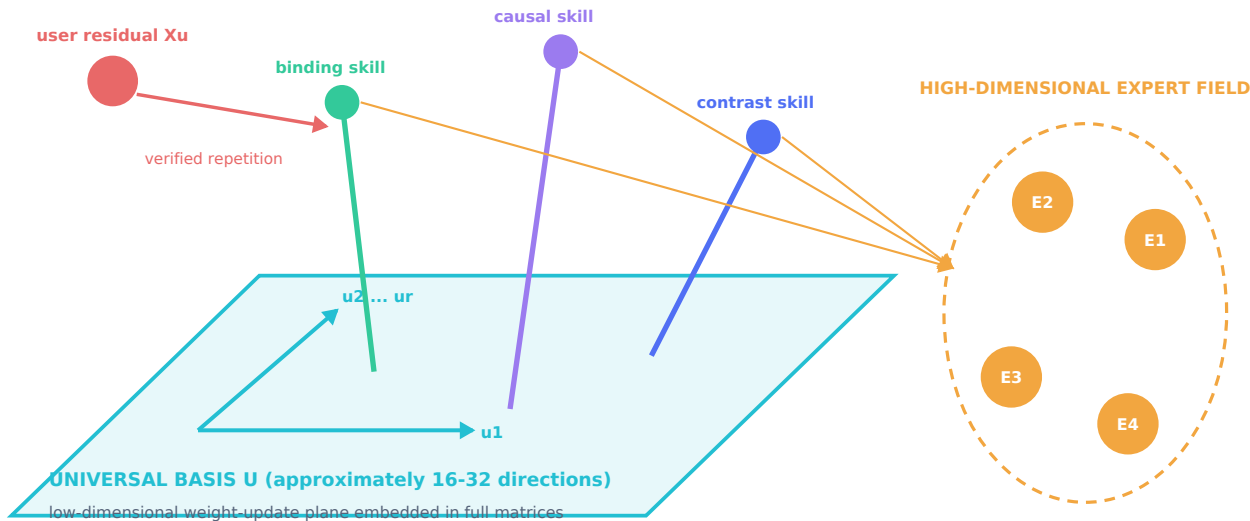
Level	Selection rule	Output
Task	Requested relation: derive, explain, compare, predict, revise	Task anchor and target roles
Skills	Euclidean geometric routing selects enough shared skill anchors to explain the task/context variance; the count may be 6, 16, 23, or another sparse amount.	LoRA operation mixture
Experts	Skill adjacency opens a bounded local candidate set; residual concept/task evidence selects the useful high-dimensional junctions.	Specialized state and branch priors
Branches	Each selected expert binds roles and constructs typed support/opposition pools.	Candidate movement, evidence, residual, trace
DeepLoop	Block 1 expands candidates; Block 2 contracts them; h returns for the next round.	Improved recurrent thought state

ROUTING ORDER

Geometric retrieval searches the compact skill anchors first. Expert retrieval then follows skill adjacency. Hundreds of experts may share one skill while each expert retains its own knowledge/task junction and branches.

8. Universal subspace, shared skill LoRAs, user LoRAs, and experts

The hierarchy lives in weight-update and operation space. The universal subspace is the smallest, slowest-changing foundation. Shared LoRA adapters are named skills that extend and reuse that foundation. A user LoRA learns fast coefficients and a temporary residual. Experts remain high-dimensional modules outside the low-rank basis.



Component	Geometry	Learning clock
Universal U	Approximately 16-32 principal directions spanning full weight-update matrices; rank follows an explained-variance threshold.	Cross-task consolidation; slowest
Shared skill L_k	A reusable LoRA operation: $L_k = UC_k + X_k$. X_k carries useful structure beyond U.	Repeated verified task evidence
User LoRA	Sparse skill coefficients plus temporary X_u residual for a user/episode.	Per visit or per episode; fastest
Expert E_e	High-dimensional knowledge + concept + task + skill junction with persistent branch specifications.	Formation and refinement after verified recurrence

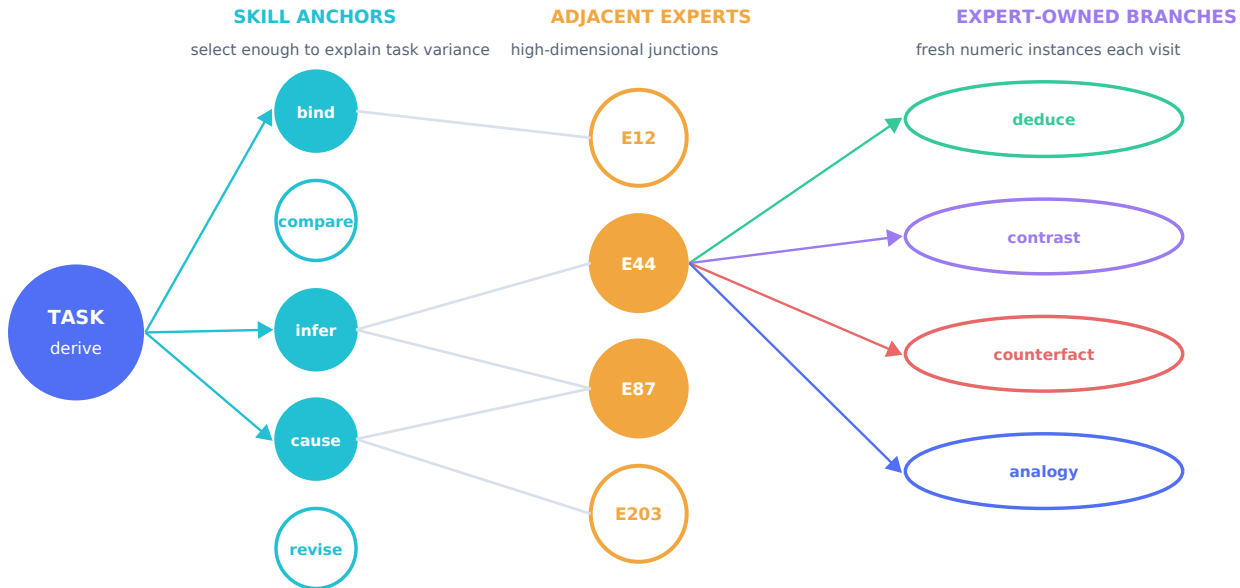
$$\Delta W_{u,e} = \sum_i c_i U_i + \sum_k a_{u,k} L_k + X_u + E_e$$

During one task, U and the shared L_k remain stable while the small coefficients $a_{u,k}$ and X_u learn.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Share [5] extracts shared principal LoRA factors, trains lightweight coefficients, preserves residual directions, and merges recurring structure through SVD. The shared-valence study [6] supplies domain-specific evidence that independently trained models can converge on a compact common direction.	Carson separates a broadly shared universal basis from named operational skill LoRAs, lets user episodes add fast coefficients and residuals, and keeps experts as high-dimensional junctions that consume combinations of skills.

9. What high-dimensional experts and their branches do

An expert is a persistent specialization junction. It binds active memory, JTD concepts, the requested task relation, selected skill adapters, prior user coefficients, and branch specifications. At each block visit, the expert creates fresh numeric branch instances from the latest thought.



Branch	Pool and operation	Primary result
Deductive	Premises, candidate conclusion, contradiction anchors	Jointly supported conclusion
Causal	Cause, mechanism, ordered effect, chronology	Mechanism-consistent progression
Contrastive	Supported candidate plus explicit alternatives	Sharper distinction and retained rejections
Counterfactual	Base pool compared with an intervention pool	Derivative/residual change caused by intervention
Abductive	Observation plus locally retrieved explanations	Best explanation with residual uncertainty
Analogical	Source relation and target anchors through shared skills	Relation-preserving transfer

Branch return contract: $B_{e,j}(h) \rightarrow (\Delta_{e,j}, \rho_{e,j}, q_{e,j}, P_{i_{e,j}})$. The tuple contains movement, unresolved harmonic residual, signed evidence, and exact source trace. Expert and block somas combine locally active returns with signed L1 normalization.

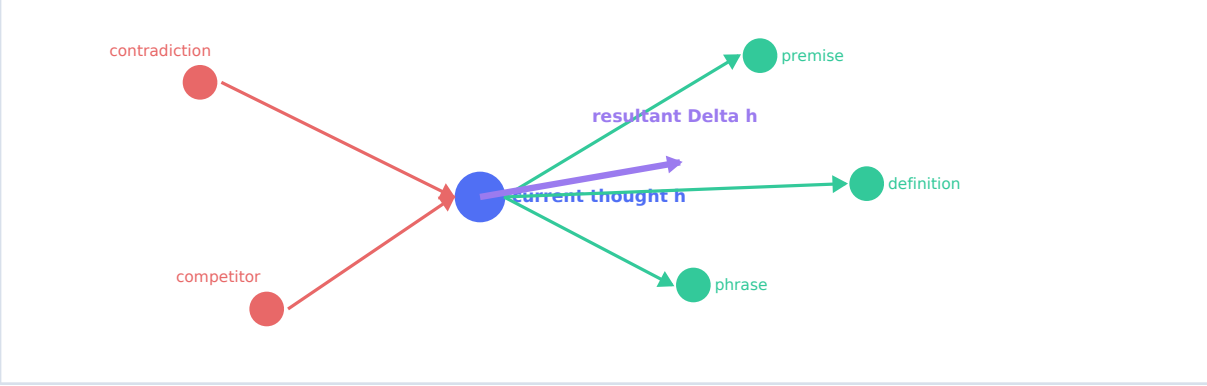
RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Bilevel NAS [12] separates outer architecture decisions from inner weight learning. Similarity-guided curriculum [13], graph-augmented evolution [14], and lineage-aware evolutionary design [15] provide adjacent strategies for progressive adapter/topology formation from measured outcomes.	These methods belong to the future expert-formation layer: an outer process may test expert adjacency and branch topology while the inner Dendritron learns coefficients. The current defining milestone is one fully executable typed deductive branch.

10. Geometric attention: Euclidean HarMax attraction and repulsion

Geometric attention acts on a bounded pool of causally valid hidden states, phrase memories, definition neighbors, LNGram concepts, and expert-branch anchors. Every runtime score is built from Euclidean distance. HarMax converts inverse-distance mass into signed movement toward underrepresented support and away from excess competing mass.

SIGNED HARMAX FIELD

green mass attracts; red mass repels; every anchor retains a source pointer



$d_i^2 = ||a_i-h||_2^2 + \text{epsilon}^2 \quad | \quad p_i = d_i^{-nu} / \sum_j d_j^{-nu} \quad | \quad c_i = y_i \cdot p_i$
 $\Delta h = nu \sum_i c_i (a_i-h) / d_i^2$

Signal	Interpretation
$c_i > 0$	Support is underrepresented; the state moves toward anchor i.
$c_i < 0$	A competitor or contradiction has excess mass; the state moves away.
$\rho = -\sum y \log p$	Residual uncertainty: the selected pool's current explanatory mismatch.

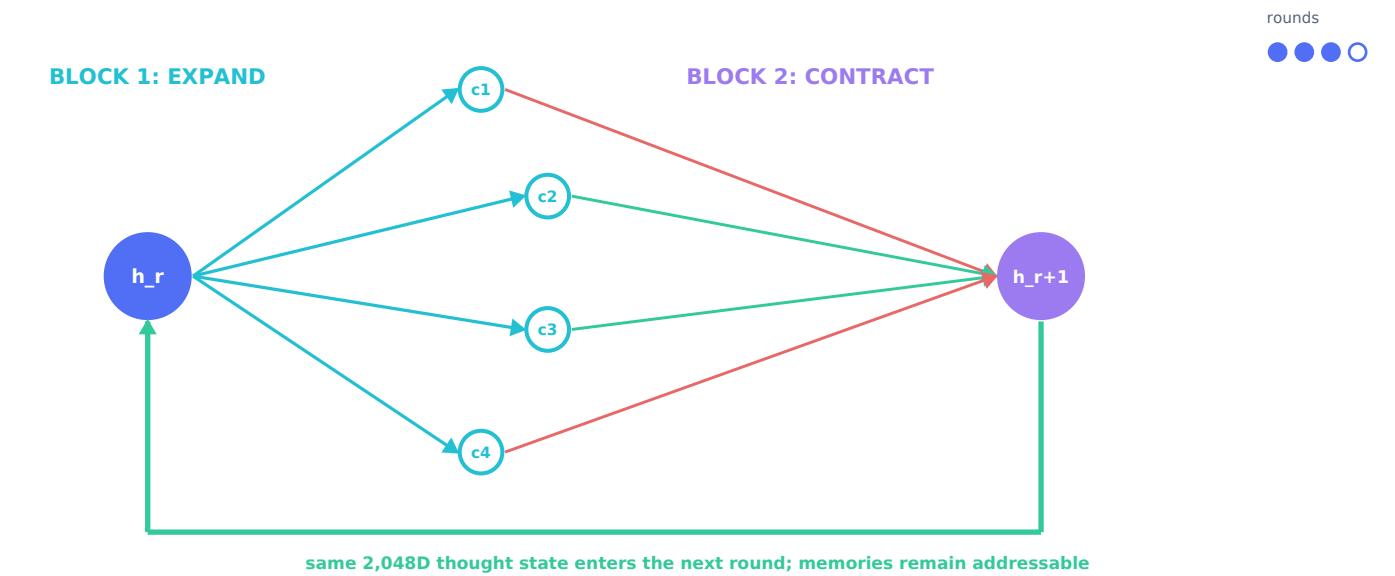
GEOMETRY LOCK

Bilinear runtime operators: 0. LNGram fusion, skill routing, concept retrieval, HarMax fields, and vocabulary ranking use Euclidean geometry.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Harmonic Loss [8] supplies inverse-distance class mass and an interpretable derivative around finite centers. Bujack et al. [10] show that global perceptual distance can display diminishing returns, motivating careful empirical tests of large-scale geometry.	Carson adapts the HarMax derivative into signed local contraction pools and begins with explicit Euclidean neighborhoods. The research program measures locality, residuals, and long-range behavior directly rather than assuming one global manifold law.

11. Where DeepLoop goes and how the two blocks carry thought

DeepLoop is the repeated execution of the two physical blocks containing the full active path. It surrounds memory influence, skill selection, expert activation, and branch testing. It therefore sits around the reasoning engine rather than after the branches.



Visit	Residual sublayer 1	Residual sublayer 2	Bias
Block 1	HarMax context, Engram/JTD memory, LNGram, branch pool	Universal/skill LoRA plus selected expert soma	Expand associations and candidate branches
Block 2	HarMax support/opposition and contradiction testing	Universal/skill LoRA plus expert integration	Contract and verify the supported state

$$h_{r+1} = B_2(B_1(h_r; m_t, U, L, E, B))$$

$K = 2$ physical blocks

$N = KR = 2R$ effective block depth

$M = 2N = 4R$ residual-sublayer visits

$\alpha = \sqrt{2N}$ on each Post-RMSNorm skip path
initialization gain

$\beta = (8N)^{-1/2}$ as one-time residual-matrix

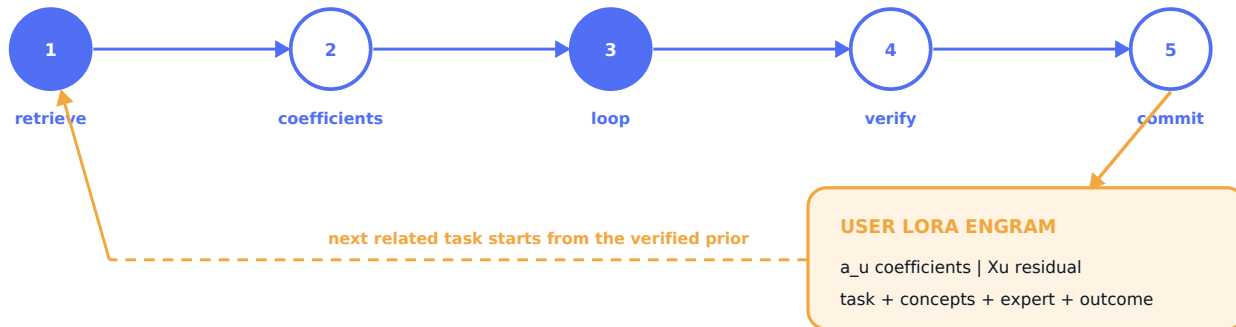
- The 2,048D activation h_r carries the evolving line of thought.
- Retrieved memory values and their pointers remain addressable at every visit; gates can refresh their influence from the latest state.
- Branch numeric activations are rebuilt from h_r , while the expert's typed branch specifications persist.
- A fixed R supplies the first implementation. Residual and convergence signals can later support adaptive test-time depth.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
DeepLoop [7] analyzes repeated physical blocks, effective unrolled depth, two residual sublayers per block, Post-RMSNorm, and loop-aware alpha/beta scaling under tied parameter visits.	Carson fixes $K=2$ and gives the blocks complementary functions: Block 1 expands semantic candidates and Block 2 contracts them through signed evidence while the same hidden state continues across rounds.

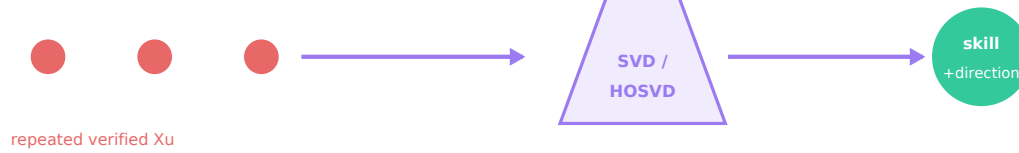
12. How user LoRA Engrams add experience and grow skills

User learning operates on a fast plastic state while the shared basis remains stable during the episode. A verified outcome becomes an experience Engram that can initialize the next related task. Repeated coherent residuals later add useful structure to a shared skill adapter.

FAST EPISODE CLOCK



SLOW CONSOLIDATION CLOCK



- **Retrieve.** User ID, task relation, LINGram concepts, selected skills, and adjacent expert IDs query the experience tier.
- **Instantiate.** The retrieved record initializes sparse coefficients $a_{u,k}$ over universal and shared skill directions plus an optional X_u residual.
- **Learn in the loop.** Per-visit or per-episode signals may include next-token error, branch consistency, answer verification, convergence energy, and direct user correction.
- **Verify and commit.** A successful episode stores coefficients, X_u , task/concept/skill/expert/branch path, outcome, residuals, and exact sources.
- **Reuse or form an expert.** Familiar interactions reuse an expert prior; recurring novel cross-interactions create a new high-dimensional junction and graph edges.
- **Consolidate.** Repeated verified X_u directions update or extend a shared skill LoRA through SVD/HOSVD; stored user coefficients are reprojected into the revised basis.

Experience record: {user, task, concept_ids, skill_coeffs, X_u , expert_id, branch_trace, sources, outcome, version}

THREE LEARNING CLOCKS

The shared skill matrices stay stable during the user's task; the user's coefficients and residual learn. Shared skills grow only after repeated verified episodes. The universal basis changes on the broadest and slowest consolidation clock.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Share [5] supports frozen principal factors, lightweight task coefficients, temporary residual factors, analytical reprojection, and periodic basis merging.	Carson stores the fast adapter state as addressable user experience, connects it to concepts and experts, and promotes repeated verified residuals into reusable named skills rather than accumulating one full adapter per task.

13. The fixed 25/75 law and CPU/ARM execution

The 25/75 split is a Dendritron design invariant over scalable conditional capacity. One quarter stores stable addressable memory; three quarters supplies conditional skills, experts, and branch computation. The two shared recurrent blocks form the fixed active substrate and remain outside this sparse-capacity ledger.



The two shared recurrent blocks are the fixed active substrate and are reported separately.

Memory - 25%	Conditional compute - 75%
Frozen 2/3-gram layer-8/24 banks	Universal operation directions
Layer-2 definition senses	Shared skill LoRA adapters
Hash-Engram and LNGram rows	High-dimensional expert modules
User experience Engrams	Typed branch executors and somas

Why Carson fixes the ratio

- The conservation split uses amplitudes 1/2 and sqrt(3)/2; their squared magnitudes are 1/4 and 3/4.
- Memory is the stable reference term; changing computation receives the threefold larger conditional region.
- Engram and LNGram provide independent empirical corroboration near a 1:3 memory-to-compute allocation.

Why the live path targets CPU/ARM

- Frozen BF16 rows are memory-mapped and fetched sparsely from CPU memory or storage.
- Exact tuples, integer hashes, and compact LNGram codes create predictable address paths.
- Skill-first graph routing activates a bounded expert/branch neighborhood.
- Two physical blocks are stored once and revisited for additional thought depth.
- User learning updates small coefficient vectors and residuals.
- Euclidean distance, chunked vocabulary ranking, quantization, and pruning map to vectorized CPU/ARM kernels.

RESEARCH BASIS	DENDRITRON-ARM EXTENSION
Engram [1] and Memory Grafting [2] use a 1:3 memory/compute allocation and demonstrate deterministic host-side retrieval. LNGram [3] reports its strongest tested allocation at 25/75. Hardware-aware GNN pruning [11] and A*-guided batch selection [16] supply adjacent precedents for target-aware quantization/pruning and compute-conscious training schedules.	Carson treats 25/75 as an architectural law whose origin is the project's own conservation and memory/compute geometry. The live target moves the complete recipient onto CPU/ARM; donor GPU work remains a one-time frozen-asset construction stage.

14. One complete end-to-end example

Prompt: 'The river bank eroded after heavy rain. Explain why.' This example exercises exact phrase memory, polysemous definition senses, JTD concept neighbors, causal skills, a specialized expert, competing branches, and recurrent verification.

Stage	Execution	Trace retained
1. Surface	Raw Qwen tokens and word boundaries preserve the exact sentence. The suffix address checks 'the river bank' then 'river bank'.	Raw spans and canonical tuples
2. Memory	A trigram or bigram hit returns layer-8 and layer-24 phrase states. The word bank exposes finance, river-edge, and tilt senses from layer 2.	Phrase rows and sense rows
3. Concept	LNGram assigns the live context a latent cell. JTD maps h, e8, and e24 into the definition frame; Euclidean neighbors favor river-edge, erosion, rainfall, and runoff.	LNGram code, JTD maps, kNN distances
4. Skills	The explain task activates causal-ordering, mechanism-binding, temporal, comparison, and contradiction-checking skill LoRAs.	Skill IDs, coefficients, route mass
5. Expert	Skill adjacency opens earth-science/hydrology experts. Residual concept evidence selects the expert joining rainfall, runoff, soil, shear, and erosion.	Expert ID and user prior
6. Block 1	Causal branches propose heavy rain -> runoff -> shear stress -> sediment removal and competing explanations such as wind or ordinary bank traffic.	Role bindings and candidate pools
7. Block 2	Chronology, definition anchors, and phrase evidence attract the runoff/erosion chain and repel finance/traffic and unrelated alternatives.	Delta, rho, signed evidence, sources
8. Loop/decode	Residual uncertainty determines another round or Euclidean vocabulary ranking. A verified user correction can commit new coefficients to the experience Engram.	Round count, token ranks, optional user commit

```
task = explain_cause
concept_neighbors = [river_bank, rainfall, runoff, shear, erosion]
skills = [causal_order, mechanism_bind, temporal_check, contradiction_check]
expert = hydrology_junction
branches = [runoff_erosion, wind_erosion, finance_bank]
return = [movement, residual, signed_evidence, exact_trace]
```

MECHANISM IN ONE SENTENCE

Exact memory supplies known semantic structure. JTD resolves the concept neighborhood. Skills supply reusable operations. The expert binds the domain-specific interaction. Branches expose alternatives. DeepLoop carries and improves the thought.

15. Current implementation boundary and next build milestone

The repository contains a runnable CPU structural core plus offline asset pipelines. The table distinguishes measured implementation from target architecture so a technical reviewer can evaluate the work precisely.

Subsystem	Current evidence	Status
Phrase inventories	500k bigram + 500k trigram inventories and layer-8/24 extraction reported complete; large tables remain external.	Built assets
Definition bank	Source inventory and resumable layer-2 extraction pipelines exist.	Pipeline coded
Surface / collisions	Canonical addressing, punctuation barriers, exact tuple verification, fingerprints, and synthetic tests exist.	Structural path
JTD	Projection classes and fit objective exist; real phrase-definition anchor fitting remains the data run.	Fit pending
Hash / LNGram / HarMax	Reference modules execute in the recorded CPU tensor suite with Euclidean geometry.	CPU-tested core
DeepLoop core	Two physical blocks, recurrence, gradients, scaling initialization, save/reload, and causal masking execute.	CPU-tested core
U / skills / experts	Generic trainable pathways execute; final fitted directions and graph population depend on real training data.	Structural path
Typed branches	Specifications and return contract exist. Six semantic executors remain the defining runtime work.	Next build
User LoRA Engrams	Coefficient retrieval, per-visit updates, verified commits, expert formation, and consolidation remain target modules.	Target runtime

Next architecture-defining milestone

Build one complete deductive executor end to end: **premise pointers -> role binding -> candidate conclusions -> support and contradiction anchors -> HarMax pool -> signed movement -> residual/confidence -> conclusion plus exact trace**. Block 1 should generate several candidates; Block 2 should verify and contract them. The same return contract can then host causal, contrastive, counterfactual, abductive, and analogical branches.

EVIDENCE BOUNDARY

The structural core proves that the data path can execute and learn on CPU. The next scientific proof is semantic: typed branches, real JTD anchors, user experience, and full training must improve held-out language and reasoning at the intended ARM cost.

16. Core references placed by mechanism

Each source below already appears in the section it supports. This page preserves complete citation identity and the precise scope used in Dendritron-ARM.

- [1] Xin Cheng et al. *Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models*. arXiv:2601.07372, 2026. Sections 1, 3, 4, 6, and 13: deterministic 2/3-gram memory, multi-head hashing, gating, allocation, and host prefetch.
- [2] Runxi Cheng et al. *Memory Grafting: Scaling Language Model Pre-training via Offline Conditional Memory*. arXiv:2605.20948, 2026. Sections 1, 3, 4, and 13: frozen donor hidden states, final-token values, exact longest suffix lookup, Qwen layers 8/24, gates, and hash fallback.
- [3] Yunao Zheng et al. *Lngram: N-gram Conditional Memory in Latent Space*. arXiv:2605.24869, 2026. Sections 4, 5, and 13: learned binary route symbols, exact latent 2/3-gram retrieval, and sparse allocation.
- [4] Jingjing Li et al. *Locality Preserving Joint Transfer for Domain Adaptation*. arXiv:1906.07441, 2019. Section 5: domain-specific projections, shared frame, and k-neighbor locality preservation.
- [5] Prakhar Kaushik et al. *Shared LoRA Subspaces for Almost Strict Continual Learning*. arXiv:2602.06043, 2026. Sections 8 and 12: principal adapter factors, lightweight coefficients, temporary residual directions, reprojection, and consolidation.
- [6] Yousef A. Radwan et al. *A Shared Valence Axis Across Modern LLMs and Human EEG: The Saturation Regularity*. arXiv:2606.00129, 2026. Section 8: domain-specific evidence for convergence onto a compact common direction and value in residual diversity.
- [7] Shuzhen Li et al. *DeepLoop: Depth Scaling for Looped Transformers*. arXiv:2607.13491, 2026. Section 11: repeated physical blocks, effective depth, two residual sublayers, Post-RMSNorm, and tied-visit scaling.
- [8] David D. Baek, Ziming Liu, Riya Tyagi, and Max Tegmark. *Harmonic Loss Trains Interpretable AI Models*. arXiv:2502.01628, 2025. Section 10: inverse-distance mass and interpretable derivative around finite centers.
- [9] Tao Lin. *A Collision-Free Hot-Tier Extension for Engram-Style Conditional Memory: A Controlled Study of Training Dynamics*. arXiv:2601.16531, 2026. Section 6: MPHf hot tier, fingerprints, multi-head cold tier, route-stratified loss, and gating dynamics.
- [10] Roxana Bujack et al. *The Non-Riemannian Nature of Perceptual Color Space*. PNAS 119(18), e2119753119, 2022. Section 10: empirical caution about global metric additivity and diminishing returns.
- [11] Piotr Wzorek, Kamil Jeziorek, and Tomasz Kryjak. *Hardware-aware Graph Neural Networks Pruning for Embedded Event-based Vision*. arXiv:2607.06739, 2026. Section 13: target-aware pruning and quantization for embedded resource constraints.

CONTRIBUTION BOUNDARY

The sources validate individual mechanisms. The integrated architecture - fixed layer-2 concept anchors, LNgram-to-JTD resolution, universal-to-skill-to-expert separation, user LoRA Engrams, typed branches, complementary two-block roles, and fixed 25/75 law - is Carson's synthesis.

17. Adjacent research sources and compact terminology

These sources belong to expert formation, training policy, or formal verification. They provide methods for later experiments while the core Dendritron mechanism remains defined by Sections 1-15.

Source	Placement in the research program
[12] Bilevel Optimization for NAS	Outer expert/branch topology decisions paired with inner weight and coefficient learning.
[13] Similarity-Guided Curriculum Fine-Tuning	Progressive skill-adaptor curriculum and adapter-merge stability studies.
[14] Graph-Augmented Evolution	Graph-aware candidate selection and measured lineage for expert formation.
[15] Evolutionary LLMs for PINN Design	Structured candidate specifications, lineage, fingerprint deduplication, and outcome-driven evolution.
[16] A*-Inspired Batch Selection	Compute-conscious curriculum scheduling for informative training examples.
[17] Extinction Depth and q-ary Codes	Finite-state analogy for tracking unresolved symbolic collisions across sequence extensions.

[12] Abhishek Shukla, Ankur Sinha, and Faiz Hamid. *Bilevel Optimization for Neural Architecture Search*. arXiv:2606.29582, 2026.

[13] Anujaya Vijayakumar, Radu Timofte, and Dmitry Ignatov. *Similarity-Guided Curriculum Fine-Tuning of LLMs for Neural Architecture Synthesis*. arXiv:2607.11591, 2026.

[14] Xuanzhou Chen and Taoli Cheng. *GAE: Graph-Augmented Evolution for Scientific Discovery via Reinforcement Optimization*. arXiv:2607.10127, 2026.

[15] Xu Yang et al. *Evolutionary Algorithm-Guided LLMs for Physics-Informed Neural Network Design*. arXiv:2607.15560, 2026.

[16] Anxhelo Shehu, Enes Stastoli, and Arben Cela. *Learning Faster without Deeper Networks: A*-Inspired Batch Selection for Efficient CNN Training*. arXiv:2607.15745, 2026.

[17] Noam Ben Shimon and Aryeh Lev Zabokritskiy (Yohananov). *Extinction Depth and q-ary Error-Correcting Codes for the Limited Permutation Channel*. arXiv:2607.19566, 2026.

Compact terminology

Term	Meaning
Engram	Sparse conditional memory addressed by exact or hashed surface patterns.
LNGram	Latent conditional memory addressed by discrete symbols learned from hidden states.
JTD	Maps phrase/live views into fixed layer-2 definition geometry for Euclidean concept neighbors.
Universal U	Approximately 16-32 principal directions in LoRA weight-update space.
Skill LoRA	Reusable learnable operation adapter shared by tasks and experts.
Expert	High-dimensional concept/task/skill junction that owns branch specifications.
Branch	Typed traceable reasoning alternative instantiated from the current thought.
DeepLoop	Block 1 expansion plus Block 2 contraction repeated for R rounds.

Exact memory retrieves stable knowledge; JTD finds its concept neighborhood; the universal basis supplies common directions; shared LoRA skills supply reusable operations; experts bind those operations to specialized context; branches expose competing inferences; and DeepLoop carries the thought until signed evidence contracts it into an answer.